

Assignment 1: Running your First Algorithm

9/8/26

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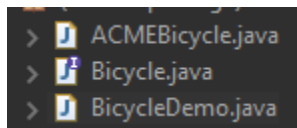
Details

Task 1) Review Oracle Interface tutorial

<https://docs.oracle.com/javase/tutorial/java/concepts/interface.html>

Deliverable: confirm finishing the tutorial and submit any code you wrote.

See



Task 2) Prepare and Clone the thinkdatastructures repo.

This includes a few tasks that you might have done before:

1-Do you have eclipse? (If not install eclipse or any JAVA IDE) <https://www.eclipse.org/downloads/>

2-Do you have a github user? If not create a github user!
<https://www.youtube.com/watch?v=w3jLJU7DT5E>

3- the code examples are stored in

<http://thinkdast.com/repo>.

4- fork the repository (what is a repository)? (from the fork button)

5- clone the repository (shown in lab)

(note: if you have not taken SENG210 yet, you can skip the step and just copy the code for each example we cover from the source (src) folder manually. So you can jump to step 6)

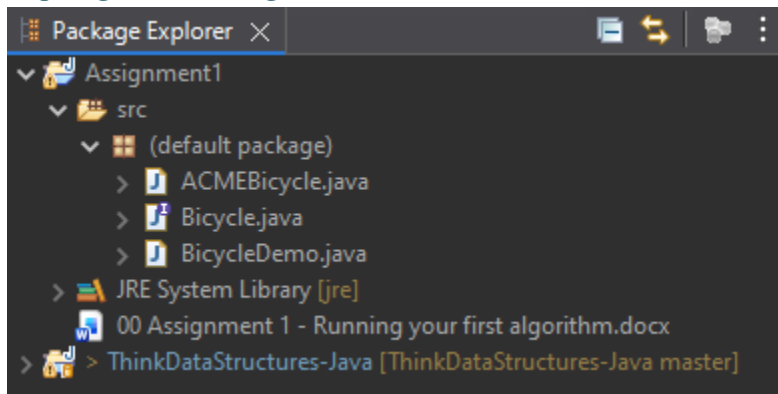
A- access Git perspective (from eclipse: window\perspective\open perspective\other\Git)

B- clone a new repo (under git repositories view-->clone new repo button --> copy your clone link taken from github)

C- Import new project from repo. (right click thinkdatastructures cloned repo, import project)

Deliverable: link to forked repository

<https://github.com/cgrimm25/ThinkDataStructures-Java>

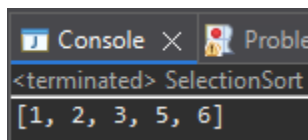


Task 3) for simplicity, we will test SelectionSort.java as a separate project.

1-create a new project and copy the SelectionSort.java code

2- test the code by running the project

Deliverable: screen shot your project output



3- Update the code to read the array contents from a text file. The size limit expected is less than 100,000.

4- Measure the performance of selection sort using the library of your choice. Basically, you are looking to measure how long does the code take to run (sort the array). one example can be found here

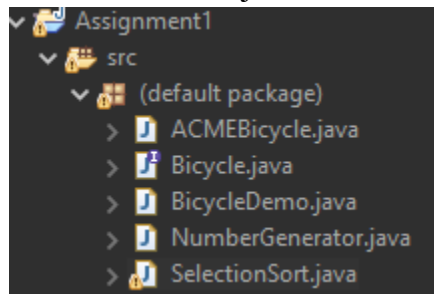
<http://tutorials.jenkov.com/java-date-time/system-currenttimemillis.html>

<https://jenkov.com/tutorials/java-date-time/time-measurement.html>

5- Run selection sort for various array sizes (using the same file) and record the running times (hint: make sure there are not print statements included in the chunk of code you are measuring, so add any prints you need AFTER you stop measuring the time, otherwise the output time will be part of the algorithm measurement - which we do not want here - printing takes the computer much more time compared to processing)

Deliverable: Code + result table for 5 different sizes

See SelectionSort.java



Before I called NumberGenerator

Array Size (n)	Execution Time (ms)
20,000	180
40,000	663
60,000	1318
80,000	2232
100,000	3422

After I added call to NumberGenerator

I was able to shave off some time. Saving more time as I processed bigger data sets.

Array Size (n)	Execution Time (ms)
20,000	136
40,000	467
60,000	1064
80,000	1848
100,000	2797